

Supplementary Reading Material: On Dual Spaces

- 阅读提示: 无限维线性空间的对偶空间是一个含糊的概念. 通常的教材上有如下三类处理方式:
 1. (右倾) 仅考虑可构造的对偶空间, 例如 $(\mathbb{R}[x])^* \simeq \mathbb{R}[[x]]$, 但对一般情形避而不谈;
 2. (折中) 明面上不使用甚至不承认选择公理, 但在背地里默认了一些反直觉的公理: 例如默认非零空间存在非零的对偶空间, 但从不给证明 (补充: 这是似乎课堂采取的方式, 但请务必验证课堂证明之正确性, 以及是否使用了选择公理 (选择公理的等价形式: 任何子空间存在补空间));
 3. (左倾) 直接使用选择公理.

有探索欲的读者不会接受右倾的处理方式, 而选择公理不在教学范围内. 本节习题将折中派的做法进行公理化, 用一条表述简单的额外公理为教材打上恰当的补丁. 我们最终承认通常公理与额外的 HB 公理.

- (通常公理). ZF 与 DC, 也就是矩阵论和数学分析承认的东西. 特别地, DC 用于证明可数个可数集的并仍可数, 以及无限维线性空间存在无穷大的线性无关组, 其严格独立于 ZF.
- (HB 公理). 对任意的子空间 $U \hookrightarrow V$, 映射 $f: U \rightarrow W$ 的定义域总能被扩大至 V . 换言之, 总能找到 $\tilde{f}: V \rightarrow W$ 使得 $\tilde{f}|_U = f$.

可以发现, 张贤科等教材的公理体系严格等价于通常公理 + HB 公理.

若不承认 HB, 我们无法排除 Exotic Examples (见习题) 的存在性. 可使用这些例子自查是否伪证了某些命题, 类似地, 有限域的重要作用也是自查伪证与否.

习题目录: 有限维对偶空间一瞥, 不承认 HB 的后果, 承认 HB 的好处, 对偶空间的 Galois 对应, 对偶空间的所有基本结论.

Finite-Dimensional Vector Spaces and Dual Spaces

Definition. Let V be a vector space over a field $\mathbb{F}$. The dual space is defined as

$$V^* := \text{Hom}_{\mathbb{F}}(V, \mathbb{F}).$$

Example. For $\dim V = n < \infty$, with basis u_i $1 \leq i \leq n$, the dual space V^* is also n -dimensional with basis (dual basis)

$$\{f_i\}_{1 \leq i \leq n}, \quad f_i: V \rightarrow \mathbb{F}, \quad u_j \mapsto \delta_{i,j}.$$

From the perspective of bilinear forms, the pairing

$$V \times V^* \rightarrow \mathbb{F}, \quad (u, f) \mapsto f(u)$$

has matrix form $I_{n \times n}$ under the bases $\{u_i\}_{1 \leq i \leq n}$ and $\{f_i\}_{1 \leq i \leq n}$.

Note. The isomorphism $V \simeq V^*$ is artificial. An informal but illustrative analogy: assume V has unit kilometre (km), then V^* has unit km^{-1} . The linear isomorphism $V \simeq V^*$ is never natural.

Example. The canonical linear map $\eta_V : V \rightarrow V^{**}$ is a natural isomorphism.

- The term "natural" signifies that for any linear map $f : U \rightarrow V$, the compositions $U \xrightarrow{f} V \xrightarrow{\eta_V} V^{**}$ and $U \xrightarrow{\eta_U} U^{**} \xrightarrow{f^{**}} V^{**}$ coincide.

Exotic Examples in Linear Algebra Without the Axiom of Choice

Example. ($\dim V = \infty$, yet $V^* = 0$). If the axiom of choice is not assumed, then there do exist infinite-dimensional vector spaces V with dual space $V^* = 0$. The [example](#) by H. Läuchli presents an infinite-dimensional vector space V whose proper subspaces are all finite-dimensional.

Example. (When V^* is strictly larger than V , yet $V \simeq V^{**}$). In ZF set theory with dependent choice, it takes effort to verify $\mathbb{R}[[x]]^* \simeq \mathbb{R}[x]$. However, $\mathbb{R}[[x]]$ does not possess a countable basis.

Proposition. The canonical linear map $V \rightarrow V^{**}$ is injective if and only if V is a subspace of some product space $\prod_X \mathbb{F}$. For instance, $\mathbb{F}[x]$ and $\mathbb{F}[[x]]$ both embed into their double dual spaces.

Proof. See [this article](#).

Axiom of Extension (HB)

Axiom. (Axiom of Extension, also known as the Hahn–Banach Axiom (HB)).

- For any linear subspace $U \subseteq V$, every linear map $f : U \rightarrow W$ extends to a map $\tilde{f} : V \rightarrow W$ such that $\tilde{f}|_U = f$.

Remark. The Hahn–Banach Axiom is independent of the usual axioms (ZF + DC), which are assumed in the study of tensor products of infinite-dimensional vector spaces.

Exercise. The Hahn–Banach Axiom is equivalent to the assertion that $\text{Hom}_{\mathbb{F}}(-, W)$ always maps injections to surjections.

Exercise. By HB, for any $v_1, v_2 \in V$ with $v_1 \neq v_2$, there exists $f \in V^*$ such that $f(v_1) \neq f(v_2)$. Hence, the canonical linear map $V \rightarrow V^{**}$ is always injective.

Exercise. By HB, show that for $U \hookrightarrow V$, the following map is an isomorphism

$$\frac{V^{**}}{U^{**}} \rightarrow (V/U)^{**}, \quad [x]_{U^{**}} \mapsto \pi^{**}(x),$$

where π is the natural quotient map $V \twoheadrightarrow V/U$.

Dual Spaces and Galois Correspondence

Definition. (Kernels and Annulateurs). For convenience, let U_i denote subspaces of V , and B_i subspaces of V^* .

- $\text{ann } U := \{f \in V^* \mid f(U) = 0\}$, a subspace of V^* ;
- $\ker B := \{u \in V \mid f(u) = 0 \ (\forall f \in B)\} = \bigcap_{f \in B} \ker f$, a subspace of V .

These notions are defined for subsets in general. By the closure property:

- $\text{ann}(X) = \text{ann}(\text{span}(X))$, for $X \subseteq V$;
- $\ker(S) = \ker(\text{span}(S))$, for $S \subseteq V^*$.

Remark. $\ker f$ and $\ker f$ are identical.

Exercise. Show that for any finite-dimensional vector space V , the map $\text{ann} : \text{Subspaces of } V \rightarrow \text{Subspaces of } V^*$ is a bijection with inverse map $\ker$.

Proposition. One has $\text{ann}(\ker B) \supseteq B$. Equality does not generally hold.

Proof. The formula $\text{ann}(\ker B) \supseteq B$ involves logic more than algebra. For a counterexample, take V as the space of continuous functions over $\mathbb{R}$, and $B = \text{span}(\{\delta_r\}_{r \in \mathbb{Q}})$.

Exercise. (Galois Connection)

1. $\text{ann}(\ker B) \supseteq B$,
2. $\ker(\text{ann } U) \supseteq U$,
3. $\text{ann}(\ker(\text{ann } B)) = \text{ann}(B)$,
4. $\ker(\text{ann}(\ker B)) = \ker B$.

The pair $(\ker, \text{ann})$ defines a Galois correspondence.

Example. For a set map $f : X \rightarrow Y$, the image and preimage maps (f_*, f^*) form a Galois connection between $\text{Subset}(X)$ and $\text{Subset}(Y)$.

Definition. Let $Op(\mathbb{R}^2)$ denote the set of open subsets of $\mathbb{R}^2$, and $Cl(\mathbb{R}^2)$ the set of closed subsets. Define:

1. $k : Op(\mathbb{R}^2) \rightarrow Cl(\mathbb{R}^2)$, $U \mapsto \overline{U}$ (closure);
2. $i : Cl(\mathbb{R}^2) \rightarrow Op(\mathbb{R}^2)$, $K \mapsto K^\circ$ (interior).

Exercise. Show that $ikik = ik$ and $kiki = ki$; hence, (k, i) form a Galois connection.

Exercise. Given $X \subseteq \mathbb{R}^2$, show that at most seven distinct sets (including X) can be obtained by applying k and i repeatedly.

Exercise. Show that one can obtain at most 14 distinct sets (including X itself) by applying i and $(-)^c$ repeatedly.

The Yoga of Dual Spaces

Theorem. In the following table, FD denotes finite-dimensional cases, G denotes general cases, and HB refers to cases where the Hahn-Banach Axiom is assumed.

For a linear map $\varphi : U \rightarrow V$, define the dual map as

$$\varphi^* : V^* \rightarrow U^*, \quad [V \rightarrow \mathbb{F}, v \mapsto f(v)] \mapsto [U \rightarrow \mathbb{F}, u \mapsto f(\varphi(u))].$$

No.	Formula	FD	G	HB
G1	$\text{ann}(\ker B) \stackrel{?}{=} B$	=	$\supseteq$	$\supseteq$
G2	$\ker(\text{ann } V) \stackrel{?}{=} V$	=	$\supset$	=
E1	$\text{ann}(V_1 + V_2) \stackrel{?}{=} \text{ann}(V_1) \cap \text{ann}(V_2)$	=	=	=
E2	$\text{ann}(V_1 \cap V_2) \stackrel{?}{=} \text{ann}(V_1) + \text{ann}(V_2)$	=	$\supseteq$	=
E3	$\ker(B_1 + B_2) \stackrel{?}{=} \ker B_1 \cap \ker B_2$	=	=	=
E4	$\ker(B_1 \cap B_2) = \ker B_1 + \ker B_2$	=	$\supseteq$	$\supseteq$
Z1	$\ker \varphi^* \stackrel{?}{=} \text{ann}(\text{im } \varphi)$	=	=	=
Z2	$\ker \varphi \stackrel{?}{=} \ker(\text{im } \varphi^*)$	=	$\subseteq$	=
Z3	$\text{im } \varphi^* \stackrel{?}{=} \text{ann}(\ker \varphi)$	=	$\subseteq$	=
Z3'	$\text{im } \pi^* \stackrel{?}{=} \text{ann}(\ker \pi)$, with π surj.	=	=	=
Z4	$\text{im } \varphi = \ker(\ker \varphi^*)$	=	$\subseteq$	=
Z5	If f is surj., then f^* is inj.	true	true	true
Z6	If f is inj., then f^* is surj.	true	NaN	true

No.	Formula	FD	G	HB
L1	$S^* \xrightarrow{?} V^* / \text{ann}(S)$	$\simeq$	$\leftarrow$	$\simeq$
L2	$(V/S)^* \xrightarrow{?} \text{ann}(S)$	$\simeq$	$\simeq$	$\simeq$

Remark. When the general case (G) coincides with the HB case, the proposition is provable without additional axioms. When the FD case differs from the G case, the FD part is proved using dimension arguments. Nonexamples disprove the HB part of statements G1 and E4.

- G1–G2 come from Galois connection,
- E1–E4 are elementary,
- Z1–Z6 are taken from the [textbook of 张贤科](#),
- L1–L2 are taken from the [textbook of 黎景辉等](#).